

Almanak strategies, executed through KeeperHub

Any Almanak strategy, unchanged, dry-runs, signs and broadcasts through KeeperHub, with one idempotency key per intent and a verified receipt back in Almanak's own parsers. No private key on the machine.

Prashant Thakur · `almanak-keeperhub` on PyPI · github.com/Prashant-thakur77/almanak-keeperhub

530+

executions on production, each with KeeperHub's verdict and decoded events

3 + 4

upstream PRs merged into KeeperHub, plus four in review, on four issues I filed and three from the maintainers' backlog

7

failure modes recorded on purpose: revert, cap, duplicate, crash, selector, RPC outage, stale exit

575 + 11

unit and property tests, plus eleven live against production every six hours

0

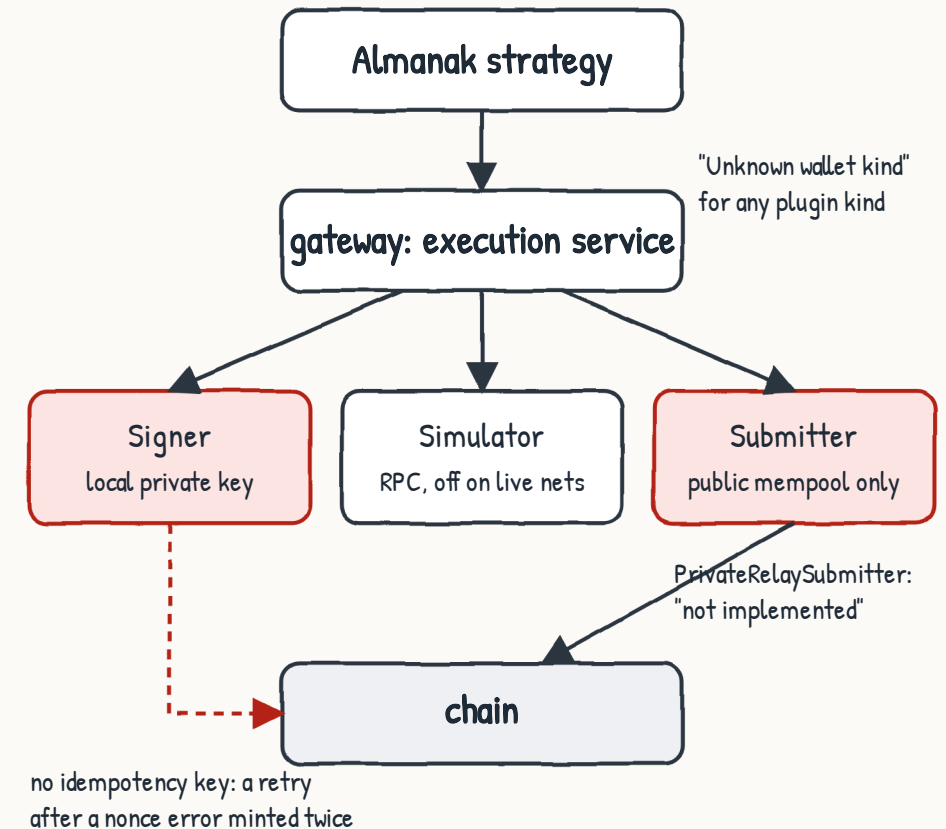
private keys on the machine, and zero cost: sponsored gas, faucet USDC

THE PROBLEM

Almanak has a clean execution layer that only knows one way to execute

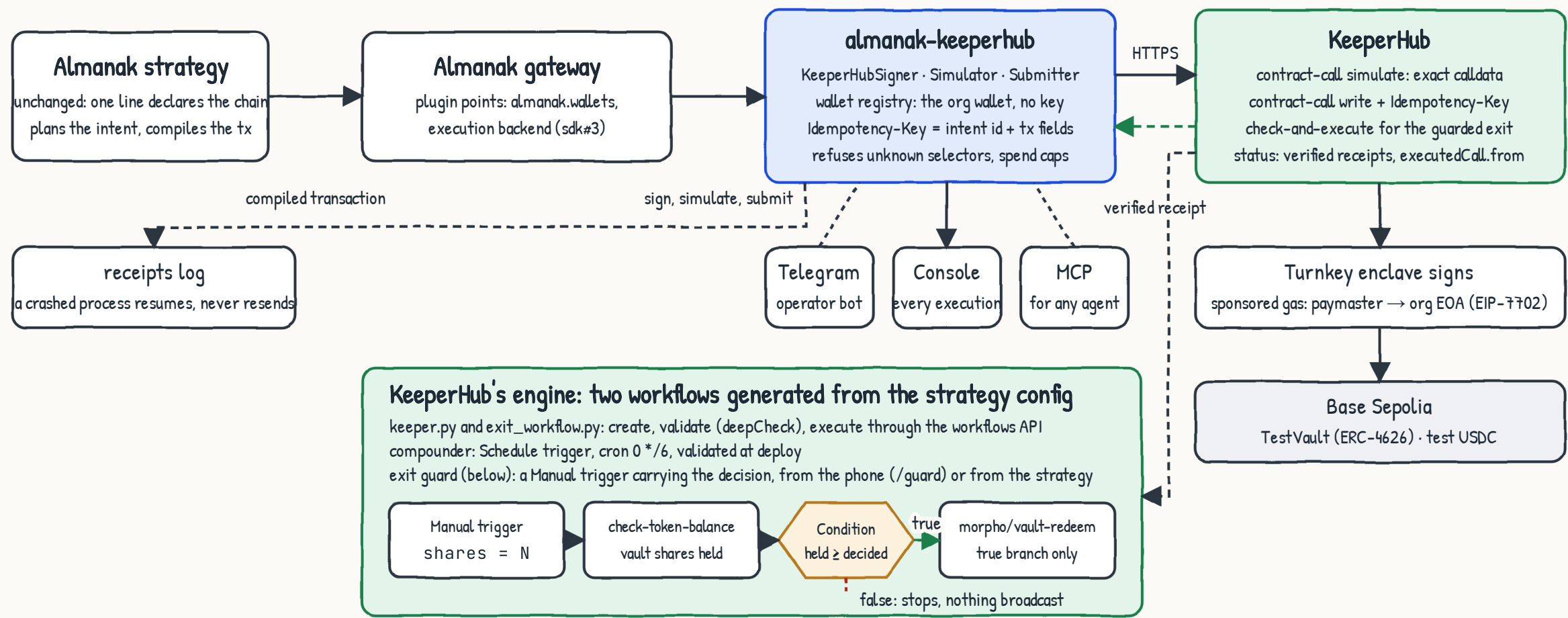
- **Almanak** is an open-source DeFi strategy framework (PyPI `almanak`, 46 protocol connectors). Its execution layer is three abstract classes: `Signer`, `Simulator`, `Submitter`.
- The only submitter that ships is the **public mempool**; the private relay is a stub that rejects every transaction.
- The **private key lives on the strategy machine**. No idempotency: their own repo carries a duplicate-mint incident after a nonce error (`nonce_recovery.py`).
- The gateway's wallet-registry plugin can **name** a wallet, but cannot say **how it signs**: an unknown wallet kind is refused, and the submitter and simulator are always the built-in ones.
- So an agent that runs an Almanak strategy has custody, retries and guards to solve by itself, exactly what KeeperHub already does.

Filed upstream with a diff that applies to their main: [almanak-co/sdk#3](#), a pluggable `almanak.execution_backends` entry-point group. Two Almanak bugs found on the way (a 30s gateway deadlock with a wallet plugin installed; simulate never enabled on live networks), both since fixed on their main.



WHAT THIS IS

KeeperHub as Almanak's execution backend: the same compiled transaction, dry-run, signed and broadcast by KeeperHub



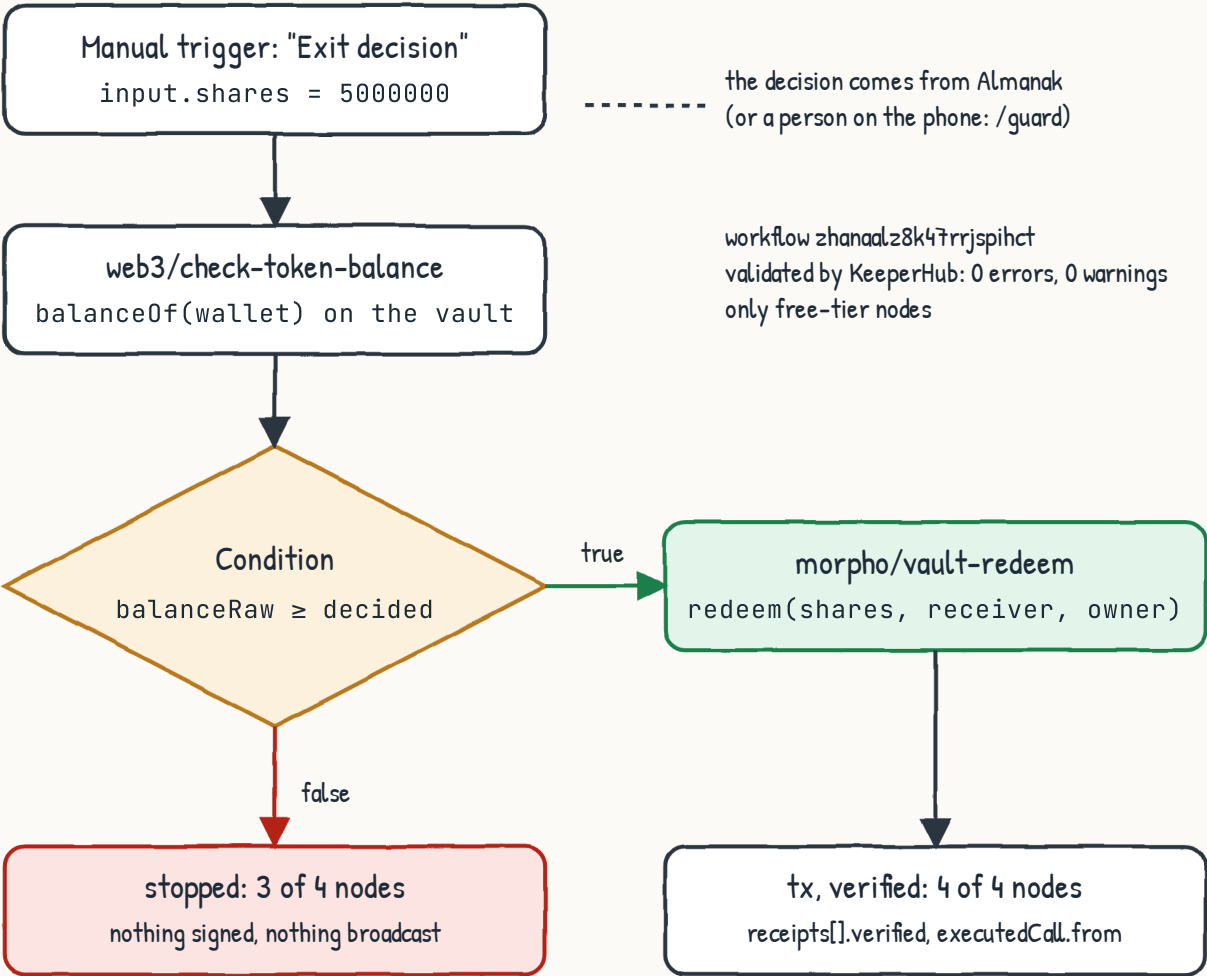
HOW IT IS PROVEN

The guard runs next to the write, and a machine repeats the proof every six hours

The guarded exit, two ways. An exit decision can be stale by the time it executes. /exit sends the redeem as KeeperHub **check-and-execute**: KeeperHub reads balanceOf(wallet) right before the write. /guard triggers the **workflow** on the right: KeeperHub's own engine reads the shares, the Condition compares, the redeem sits behind the true branch.

RUN, 22 SEP	RESULT
stale decision: 5,000,001 vs 5,000,000 held qez8b9fhipm7c4zcqa6sg	STOPPED AT THE CONDITION 3 of 4 nodes, nothing broadcast, 3.7 s
live decision: 5,000,000 90eswt00kn44cw2szpl80	REDEEMED tx 0xc18e3c7f...2479d, verified, 8.4 s

Benchmark (bench-1789496159): 50/50 impossible deposits refused before broadcast · 200/200 approvals verified, p50 7.8 s · 50/50 retries replayed by key, 0 double broadcasts · 10/10 killed processes settled by a fresh one, 0 resent. **Runner**, every six hours, nobody present: tick, guarded exit, 11 conformance tests against production, console export, committed as proof-tick.



Four gaps found in KeeperHub's API, filed, accepted, built; three merged and live

KEEPERHUB	ISSUE → PR	STATE
Raw calldata on contract-call, decoded against the verified ABI	#2426 → #2449	MERGED
Simulate a sequence, each call against the state the previous one produced (calls[])	#2427 → #2452	MERGED
The acting wallet on sponsored executions: executedCall.from	#2428 → #2450	MERGED
Workflow preflight dry-runs against carried state, capped, per node fallback	#2519 → #2531	IN REVIEW
Approve-side allowance hint (maintainers' backlog)	#2367 → #2533	IN REVIEW
Adding an EVM chain: runbook plus a seed guard (maintainers' backlog)	#2497 → #2532	IN REVIEW
Fixed-point aggregation in the math plugin (maintainers' backlog)	#2496 → #2534	IN REVIEW
Almanak: pluggable execution backend for the gateway, with a diff that applies to their main	almanak-co/sdk#3	PROPOSED

Seven pull requests in twelve days, each reviewed line by line by the same two maintainers. Two Almanak bugs reported and since fixed on their main.

NEXT

- **Mainnet.** The code path is identical; only funding was withheld. doctor and the console footer already say which chain is live.
- **Whole strategies as KeeperHub workflows.** The exit guard is the first node graph the package generates from a strategy; the compounder is the second. With #2531's preflight, every later node is dry-run against the state the earlier ones produce before anything is enabled.
- **Almanak's Safe plus Zodiac Roles mode** mapped onto KeeperHub's policy layer: roles decide what an agent may call, KeeperHub decides whether it lands.
- **Agents as operators.** The MCP server already serves the strategy with KeeperHub's read/write key split; next, the guard becomes the default shape of every agent write.
- **The Almanak side.** When almanak.execution_backends lands, the package needs no subclass at all: a registry entry of kind keeperhub is the whole integration.